Chapter 5: Predictive Analytics

Supervised Machine Learning

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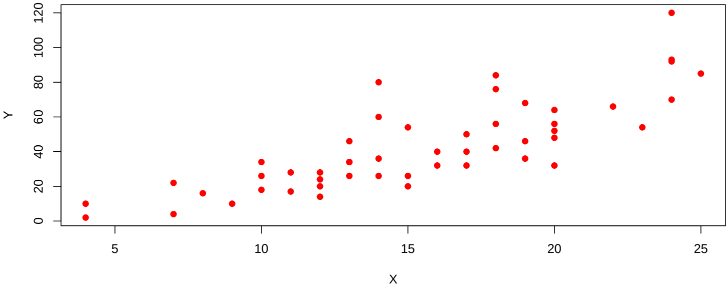
Simple Linear Regression

- The two-variable regression model assigns one of the variables the status of an *independent variable* (predictor, X), and the other variable the status of a *dependent variable* (response, Y).
- The independent variable may be regarded as causing changes in the dependent variable, or the indepen-
dent variable may occur prior in time to the dependent variable.



Simple Linear Regression

Example

A data frame with 50 observations on 2 variables.

1. *speed* numeric Speed (mph)
2. *dist* numeric Stopping distance (ft)

```r
head(cars)
```

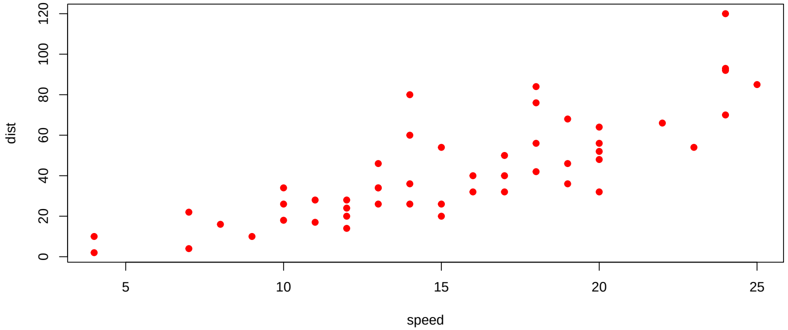
```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

Simple Linear Regression

Example

Is there a correlation between the speed and distance in the *cars* data?

```
plot(cars, pch=19, col = "red")
```



Simple Linear Regression

Correlation r

$$r = \frac{N\sum XY - (\sum X \sum Y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

```
r=(length(cars$speed)*sum(cars$speed*cars$dist)-
    (sum(cars$speed)*sum(cars$dist)))/
    (sqrt((length(cars$speed)*sum(cars$speed^2)-
    (sum(cars$speed))^2)*(length(cars$speed)*
    sum(cars$dist^2)-(sum(cars$dist))^2)))
r
```

```
[1] 0.8068949
```

Simple Linear Regression

Example

```
cor(cars)
```

```
          speed      dist
speed 1.0000000 0.8068949
dist  0.8068949 1.0000000
```

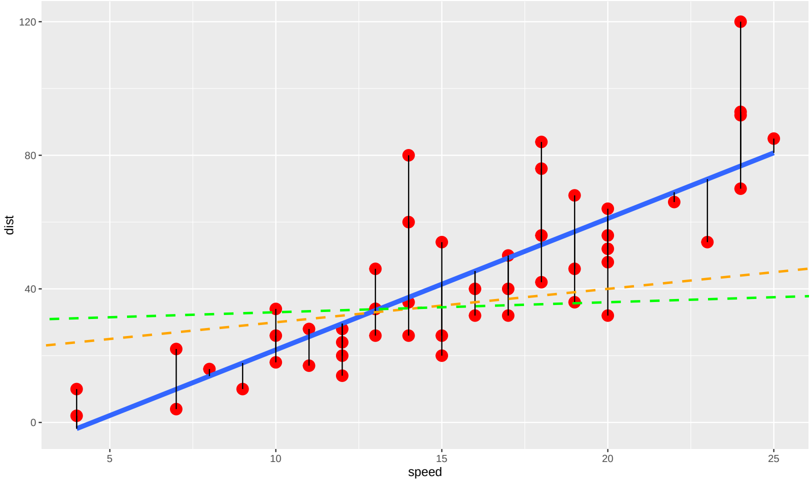
Size of Correlation r	Interpretation
0.9 to 1.0 (-0.9 to -1.0)	Very high positive (**negative**) correlation
0.7 to 0.9 (-0.7 to -0.9)	High positive (**negative**) correlation
0.5 to 0.7 (-0.5 to -0.7)	Moderate positive (**negative**) correlation
0.3 to 0.5 (-0.3 to -0.5)	Low positive (**negative**) correlation
0.0 to 0.3 (0.0 to -0.3)	Negligible correlation

Thus, the *speed* is highly positively correlated with the *stopping distance*.

Simple Linear Regression

How does it work?

It works by making the total of the square of the errors as small as possible. Why it is called "least squares".



Simple Linear Regression

The procedure most used is the *least squares criterion*, and the regression line that results from this is called the **least squares regression line**.

$$\hat{Y} \approx \beta_0 + \beta_1 X$$

- More precisely, we have the model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

  where the term ϵ is a mean-zero random error term.
- Formula for the **slope**:

$$\beta_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

- Formula for the y-**intercept**:

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

Simple Linear Regression

```
beta1=sum((cars$speed-mean(cars$speed))*
          (cars$dist-mean(cars$dist)))/
  sum((cars$speed-mean(cars$speed))^2)
beta1
```

```
[1] 3.932409
```

```
beta0=mean(cars$dist)-beta1*mean(cars$speed)
beta0
```

```
[1] -17.57909
```

Simple Linear Regression

```
beta0; beta1
```

```
[1] -17.57909

[1] 3.932409
```

```
smodel=lm(dist~speed, data = cars)
smodel
```

```
Call:
lm(formula = dist ~ speed, data = cars)

Coefficients:
(Intercept)        speed
    -17.579        3.932
```

Thus, the equation of the regression line is

$$dist = -17.579 + 3.932(speed).$$

Simple Linear Regression

$$dist = -17.579 + 3.932(speed)$$

- Interpretation of the coefficient: for every increase of 1 mph of *speed*, the stopping *distance* will increase by 3.932 ft.

R-squared

- R-squared (square of the coefficient of correlation r) is a *statistical measure of how close the data are to the fitted regression line*.
- It is also known as the coefficient of determination, or the coefficient of multiple determination for multiple regression.

Simple Linear Regression

```r
summary(smodel)
```

```
Call:
lm(formula = dist ~ speed, data = cars)

Residuals:
    Min      1Q  Median      3Q     Max
-29.069  -9.525  -2.272   9.215  43.201

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791     6.7584  -2.601   0.0123 *
speed         3.9324     0.4155   9.464 1.49e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared:  0.6511,    Adjusted R-squared:  0.6438
F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

Thus, 65.11% indicates that the model explains all the variability of the *stopping distance* variable around its mean.

Simple Linear Regression

Is the model signficant?

Next to consider:

Does the identified (linear regression) model provide a good fit for the data?

- H_o:The model is NOT significant. (i.e.,the model does not provide a good fit for the data)
- H_a:The model is significant. (i.e.,the model provides a good fit for the data)

Since the **p-value=1.49e-12** with *F=89.56711* is less than 0.05 (reject the null hypothesis), so the *model is significant* or the *model provides a good fit for the data*.

```
summary(smodel)$fstatistic
```

```
   value    numdf    dendf
89.56711  1.00000 48.00000
```

Simple Linear Regression

Does the independent variable constribute significantly?

Let β_i be the true (population) slope coefficient between *independent* and *dependent* variables. Then we wish to evaluate the following hypotheses:

- H_o: $\beta_i = 0$ (i.e., X does not contribute significantly in the model)
- H_a: $\beta_i \neq 0$ (i.e., X contributes significantly in the model)

Since the p-value=1.49e-12 is less than 0.05, so the *speed* constributes significantly in the model.

```
summary(smodel)$coefficients
```

```
             Estimate Std. Error   t value     Pr(>|t|)
(Intercept) -17.579095  6.7584402 -2.601058 1.231882e-02
speed         3.932409  0.4155128  9.463990 1.489836e-12
```

Simple Linear Regression

Standard Errors

```
Ignoring unknown labels:
* size : "10"
```

Simple Linear Regression

Standard Errors

The error is the difference between the actual value and the predicted value and it is given the symbol, ϵ. For observation i,

$$\epsilon_i = Y_i - \hat{Y}_i$$

Note: When all these errors of estimate are added, the sum is 0. That is,

$$\sum_{i=1}^{n} \epsilon_i = 0.$$

The **standard error of the estimate** is often given the symbol s_e and is

$$s_e = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}}.$$

Residual standard error: 15.38 from the summary.

Simple Linear Regression

How to Calculate the Residual Sum of Squares (RSS)

$$RSS = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

$$s_e = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}}$$

```r
RSS=sum((residuals(smodel))^2)
se=sqrt(RSS/(length(cars$speed)-2))
se
```

```
[1] 15.37959
```

```r
summary(smodel)$sigma
```

```
[1] 15.37959
```

Simple Linear Regression

Analysis of Variance (ANOVA) consists of calculations that provide information about levels of variability within a regression model and form a basis for tests of significance.

```
anova(smodel)
```

```
Analysis of Variance Table

Response: dist
          Df Sum Sq Mean Sq F value   Pr(>F)
speed      1  21186 21185.5  89.567 1.49e-12 ***
Residuals 48  11354   236.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Simple Linear Regression

How to Calculate the Total Sum of Squares (TSS)

$$TSS = ESS + RSS$$

The explained sum of squares (ESS) is the sum of the squares of the deviations of the predicted values from the mean value of a response variable (Y), in a standard regression model.

$$ESS = \sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2$$

```
ESS=sum((fitted(smodel)-mean(cars$dist))^2); ESS;RSS
```

```
[1] 21185.46
```

```
[1] 11353.52
```

```
TSS=ESS+RSS; TSS
```

```
[1] 32538.98
```

Simple Linear Regression

Predicted Value and Confidence Interval

Sample estimate $\pm$ (t-multiplier x standard error) and the formula is

$$\hat{Y} \pm t_{\alpha/2, n-2}\sqrt{MS_{Res}\left(1+\frac{1}{n}+\frac{(X_0-\bar{X})^2}{\sum(X_i-\bar{X})^2}\right)}$$

Suppose that the speed is 6.5, then the distance is:

```
-17.5791+3.9324*6.5
```

```
[1] 7.9815
```

```
smodel$coefficients
```

```
(Intercept)       speed
 -17.579095    3.932409
```

Simple Linear Regression

Predicted Value and Confidence Interval

```
new.current = data.frame(speed = 6.5)
predict (smodel, new.current, interval = "confidence",
         level = 0.95)
```

```
       fit        lwr     upr
1 7.981562 -0.6445787 16.6077
```

Using the model, if the *speed* is 6.5 mph then the *stopping distance* is approximately 7.98 feet within the interval (-0.6445787, 16.6077).

```
head(smodel$fitted.values,5)
```

```
        1         2         3         4         5 
-1.849460 -1.849460  9.947766  9.947766 13.880175 
```

Simple Linear Regression

Predicted Value and Confidence Interval

Multiple Linear Regression: Data Code

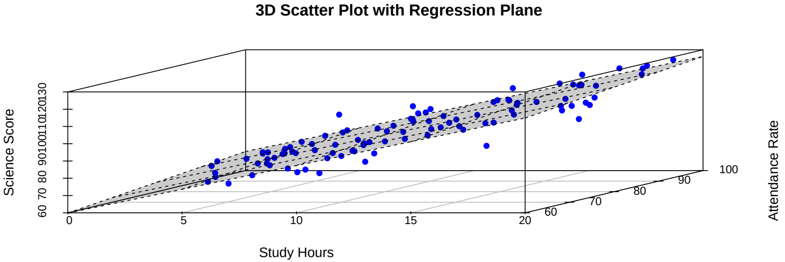
```
set.seed(123)
n <- 100
data <- data.frame(
  StudyHours = round(runif(n, 2, 20), 1),
  AttendanceRate = round(runif(n, 60, 100), 1),
  Gender = factor(sample(c("Male", "Female"), n,
                         replace = TRUE)),
  SchoolType = factor(sample(c("Public", "Private"),
                             n, replace = TRUE))
)
# Generate Science Score with some realistic relationship
data$ScienceScore <- round(
  40 +
  2.5 * data$StudyHours +
  0.3 * data$AttendanceRate +
  ifelse(data$Gender == "Female", 3, 0) +
  ifelse(data$SchoolType == "Private", 5, 0) +
  rnorm(n, 0, 5),
  1
)
```

Multiple Linear Regression

In practice we have more than one predictors (p). Instead of fitting a separate simple linear regression model for each predictor, we extend the simple linear regression model to

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon.$$

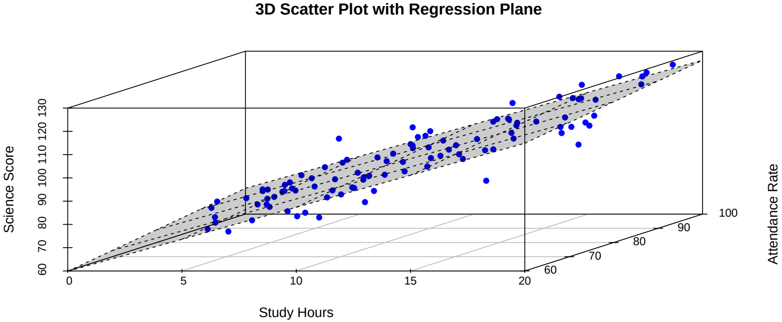
Here, we interpret β_i as the average effect on Y of a *one unit increase in* X_i*, holding all other predictors fixed*.



Multiple Linear Regression

Example

In a three-dimensional setting, with two predictors
and one response, the least squares regression line becomes a **plane**.



$$\text{Science Score} = \beta_0 + \beta_1 \cdot \text{Study Hours} + \beta_2 \cdot \text{Attendance Rate}$$

Multiple Linear Regression

TESTING FOR OVERALL SIGNIFICANCE

1. ASSESSING STATISTICAL SIGNIFICANCE

- STEP 1: Assess Model Significance (F-test on model significance)
  Does the model provide a good fit for the data?
- STEP 2: Assess Individual Predictor Significance - Proceed to this step **Only if MODEL IS SIGNIFICANT!**

2. ASSESSING MODEL SIGNIFICANCE

- The ANOVA approach applies
- COEFFICIENT OF (MULTIPLE) DETERMINATION R^2

Multiple Linear Regression

RATIONALE

- It is the extension of the simple linear regression model to account for simultaneously using multiple variables to predict a single response.

PURPOSES

- to predict a certain response, given a combination of predictor variables
- to assess the relative importance of each predictor variable in explaining the response variable Y

Design: one dependent variable (response) at either an interval or ratio measurement and several independent variables (predictors) at any measurement scale

NO CAUSATION Implied!

- Regression deals with dependence amongst variables within a model. But it cannot always imply causation. For example, we stated above that *speed* affects *stopping distance* and there is data that supports this.
- However, this is a one-way relationship: *stopping distance cannot* affect *speed*. It means that there is no cause-and-effect reaction on regression if there is no causation.
- In short, we conclude that a statistical relationship does not imply causation.

Multiple Linear Regression

Data

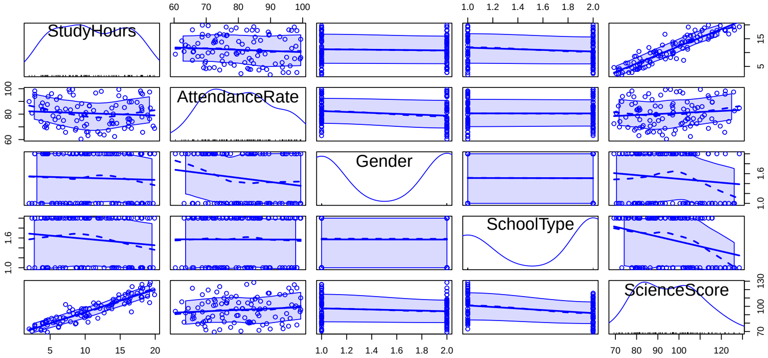
- *ScienceScore* – numeric (dependent variable)
- *StudyHours* – numeric (hours studied per week)
- *AttendanceRate* – numeric (percentage of classes attended)
- *Gender* – categorical (Male, Female)
- *SchoolType* – categorical (Public, Private)

	StudyHours	AttendanceRate	Gender	SchoolType	ScienceScore
1	7.2	84.0	Female	Private	102.2
2	16.2	73.3	Female	Private	117.1
3	9.4	79.5	Female	Public	89.0
4	17.9	98.2	Male	Private	121.9
5	18.9	79.3	Female	Public	112.0
6	2.8	95.6	Female	Public	76.3

Multiple Linear Regression

Scatter Plots

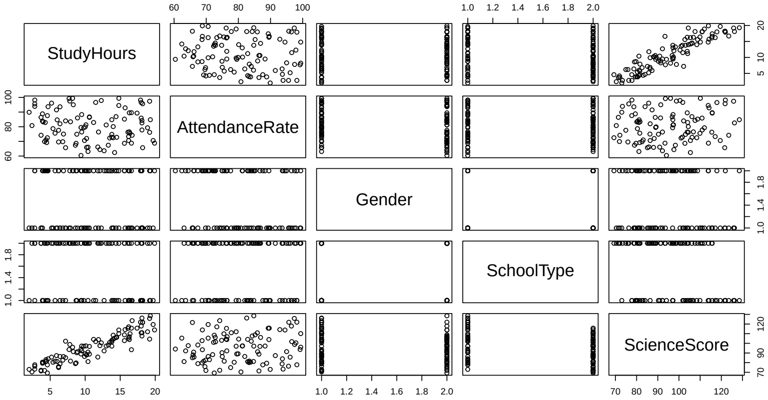
```r
library(car)
scatterplotMatrix(data)
```



Multiple Linear Regression

Scatter Plots

```r
plot(data)
```



The Model

The general multiple linear regression model with response Y and regressors $X_1, \cdots, X_p$ will have the form:

$$E(Y \mid X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

The symbol X in $E(Y \mid X)$ means that we are conditioning on all the regressors on the right side of the equation.

When we are conditioning on specific values for the predictors $x_1, \cdots, x_p$, which we will collectively call $\mathbf{x}$, we write:

$$E(Y \mid X = \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

The β's are unknown parameters to be estimated.

Data and Matrix Notation

We represent the response vector **Y** and the design matrix **X** as:

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}$$

The ith row of **X** is denoted by the symbol $\mathbf{x}_i'$, which is a $(p+1) \times 1$ vector (including the intercept).

The symbol $'$ denotes **transpose**.

Deriving the Matrix Form from the Regression Equation

Step 1: Vector Form of One Observation

We define the **row vector** of predictors for observation *i* as:

$$\mathbf{x}_i' = \begin{bmatrix} 1 & x_{i1} & x_{i2} & \cdots & x_{ip} \end{bmatrix}$$

And the **column vector** of coefficients as:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

So the predicted value becomes:

$$\hat{y}_i = \mathbf{x}_i' \boldsymbol{\beta}$$

Deriving the Matrix Form from the Regression Equation

Step 2: Stacking All Observations

We define:

- $\mathbf{Y}$: the $n \times 1$ vector of all responses,
- $\mathbf{X}$: the $n \times (p+1)$ matrix of predictors (including intercept),
- ε: the $n \times 1$ vector of errors.

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Then the **matrix form** of the linear model becomes:

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon$$

Deriving the Matrix Form from the Regression Equation

Step 3: Mean Function in Matrix Form

The conditional expectation (mean function) of $\mathbf{Y}$ given $\mathbf{X}$ is:

$$E(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\beta$$

This form is compact, efficient, and forms the basis for deriving the least squares estimator.

Goal: Estimate β

We estimate β by minimizing the **sum of squared residuals**.
The **residuals** are defined as:

$$\varepsilon = \mathbf{Y} - \mathbf{X}\beta$$

The **residual sum of squares (RSS)** is then:

$$RSS(\beta) = \varepsilon^\top \varepsilon$$

Deriving the Matrix Form from the Regression Equation

Substitute $\varepsilon = \mathbf{Y} - \mathbf{X}\beta$:

$$RSS(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta)$$

This is a **scalar** quantity (a real number), measuring the total squared difference between observed and predicted values.

Expand the RSS Expression

$$RSS(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta$$

- This form is useful for taking derivatives to find the least squares estimate $\hat{\beta}$.

We proceed by minimizing this expression with respect to β to find the best-fitting linear model.

Derivation in Matrix Form

We aim to find the least squares estimator $\hat{\beta}$ that minimizes the sum of squared residuals:

$$\text{RSS} = (\mathbf{Y} - \mathbf{X}\beta)^\top(\mathbf{Y} - \mathbf{X}\beta)$$

Step 1: Expand the Residual Sum of Squares

$$\begin{aligned} \text{RSS} &= (\mathbf{Y} - \mathbf{X}\beta)^\top(\mathbf{Y} - \mathbf{X}\beta) && (1) \\ &= \mathbf{Y}^\top\mathbf{Y} - 2\beta^\top\mathbf{X}^\top\mathbf{Y} + \beta^\top\mathbf{X}^\top\mathbf{X}\beta && (2) \end{aligned}$$

Step 2: Take the Derivative with Respect to β

$$\frac{\partial \text{RSS}}{\partial \beta} = -2\mathbf{X}^\top\mathbf{Y} + 2\mathbf{X}^\top\mathbf{X}\beta$$

Derivation in Matrix Form

Step 3: Set Derivative to Zero and Solve for $\hat{\beta}$

$$-2\mathbf{X}^\top\mathbf{Y} + 2\mathbf{X}^\top\mathbf{X}\hat{\beta} = 0$$

$$\Rightarrow \mathbf{X}^\top\mathbf{X}\hat{\beta} = \mathbf{X}^\top\mathbf{Y}$$

Step 4: Isolate $\hat{\beta}$

Assuming $\mathbf{X}^\top\mathbf{X}$ is invertible:

$$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$$

This is the **least squares solution** to the multiple linear regression problem.

Estimating Coefficients in Matrix Form

The least squares estimate of the regression coefficients β in a multiple linear regression model is given by:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Where:

- $\mathbf{X}$ is the $n \times (p + 1)$ design matrix (including the intercept),
- $\mathbf{Y}$ is the $n \times 1$ response vector,
- $\hat{\beta}$ is the $(p + 1) \times 1$ vector of estimated regression coefficients.

Example

```
X <- matrix(c(1, 5, 6,   # Stud 1: 5 hrs studied, 6 hrs slept
              1, 3, 5,   # Stud 2: 3 hrs studied, 5 hrs slept
              1, 8, 7),  # Stud 3: 8 hrs studied, 7 hrs slept
            nrow = 3, byrow = TRUE)
Y <- matrix(c(90, 70, 100), nrow = 3, byrow = TRUE)
# Compute beta hat
XtX_inv <- solve(t(X) %*% X)
XtY <- t(X) %*% Y
beta_hat <- XtX_inv %*% XtY
beta_hat
```

```
     [,1]
[1,] -100
[2,]  -10
[3,]   40
```

Example

```r
X <- matrix(c(1, 5, 6,    # Stud 1: 5 hrs studied, 6 hrs slept
              1, 3, 5,    # Stud 2: 3 hrs studied, 5 hrs slept
              1, 8, 7),   # Stud 3: 8 hrs studied, 7 hrs slept
            nrow = 3, byrow = TRUE)
Y <- matrix(c(90, 70, 100), nrow = 3, byrow = TRUE)
# Extract predictor columns from matrix X
hours_studied <- X[, 2]
hours_slept <- X[, 3]
# Response vector
test_score <- Y[, 1]
# Create data frame
df <- data.frame(test_score, hours_studied, hours_slept)
# Fit the model using lm()
model <- lm(test_score ~ hours_studied + hours_slept, data = df)
model
```

```

Call:
lm(formula = test_score ~ hours_studied + hours_slept, data = df)

Coefficients:
  (Intercept)  hours_studied    hours_slept
         -100            -10             40
```

Multiple Linear Regression

Model

```
mmodel=lm(ScienceScore~., data = data)
summary(mmodel)
```

```
Call:
lm(formula = ScienceScore ~ ., data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-11.336  -3.208  -0.671   2.925  11.753 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)      48.95167    4.08531  11.982  < 2e-16 ***
StudyHours        2.67241    0.09259  28.863  < 2e-16 ***
AttendanceRate    0.26493    0.04525   5.855 6.79e-08 ***
GenderMale       -1.37590    0.94737  -1.452     0.15    
SchoolTypePublic -5.42863    0.94930  -5.719 1.24e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.656 on 95 degrees of freedom
Multiple R-squared:  0.9092,	Adjusted R-squared:  0.9053 
F-statistic: 237.7 on 4 and 95 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression

ASSESSING STATISTICAL SIGNIFICANCE

1. With a p-value $= 2.2\text{e-}16 < \alpha$, the model is significant.
2. All the predictors are significant.

```
anova(mmodel)
```

```
Analysis of Variance Table

Response: ScienceScore
               Df  Sum Sq Mean Sq  F value    Pr(>F)
StudyHours      1 19003.1 19003.1 876.4492 < 2.2e-16 ***
AttendanceRate  1   859.9   859.9  39.6614 9.334e-09 ***
Gender          1    41.8    41.8   1.9279    0.1682
SchoolType      1   709.0   709.0  32.7018 1.239e-07 ***
Residuals      95  2059.8    21.7
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

All independent variables are significant except *Gender*.

Multiple Linear Regression

ASSESSING MODEL SIGNIFICANCE

```
summary(mmodel)$r.squared
```

```
[1] 0.9091553
```

- 85.41% indicates that the model explains all the variability of the Daily Calorie Intake variable around its mean.

```
summary(mmodel)$sigma
```

```
[1] 4.656385
```

- The residual standard error is 4.656385.

Multiple Linear Regression

ASSESSING MODEL SIGNIFICANCE

```r
options(width = 70)
mmodel$coefficients
```

```
    (Intercept)        StudyHours   AttendanceRate       GenderMale 
     48.9516658         2.6724084        0.2649308       -1.3759026 
SchoolTypePublic 
     -5.4286289 
```

Coefficient Interpretation

$$\begin{aligned} ScienceScore = 48.9516658 + 2.6724084(StudyHours) \\ + 0.2649308(AttendanceRate) - 1.3759026(GenderMale) \\ -5.4286289(SchoolTypePublic) \end{aligned}$$

- For every 1 hour *increase* in the *Study Hour*, the *Science Score* will *increase* by 2.6724084 assuming that the other predictors are fixed.
- For every 1 percent *increase* in the *Attendance Rate*, the *Science Score* will *increase* by 0.2649308 assuming that the other predictors are fixed.

Multiple Linear Regression

QUALITATIVE PREDICTORS: GENERAL CASE

```
table(data$Gender)
```

```
Female   Male
    49     51
```

Consider the following Gender variable size:

$$z = \begin{cases} \textit{Female} \\ \textit{Male} \end{cases}$$

Since z has $c = 2$ levels, then to have it as a predictor in a regression model one uses $c - 1 = 1$ dummy variables which can be defined as follows:

$$z_{\text{Female}} = \begin{cases} 1 & \text{if } z = \text{Female} \\ 0 & \text{otherwise} \end{cases} \quad \textit{and} \quad z_{\text{Male}} = \begin{cases} 1 & \text{if } z = \text{Male} \\ 0 & \text{otherwise} \end{cases}$$

The *Female* category is then used as the *baseline/reference* category for interpretation purposes.

Multiple Linear Regression

QUALITATIVE PREDICTORS: GENERAL CASE

The *lm* function will automatically convert the variables.

```
options(width = 50)
mmodel$coefficients
```

```
     (Intercept)       StudyHours
      48.9516658        2.6724084
  AttendanceRate       GenderMale
       0.2649308       -1.3759026
SchoolTypePublic
      -5.4286289
```

For a *Male* student, there is a **decrease** of *Science Score* by 1.3759026 assuming the rest of the predictors are fixed.

Multiple Linear Regression

After obtaining the least-squares fit, check for the following:

- How well does this equation fit the data? (Done!)
- Is the model likely to be useful as a predictor? (Done!)
- Are any of the basic assumptions (such as constant variance and uncorrelated errors) violated? (Not yet done!)

All of these issues must be investigated before the model is finally adopted for use.

OUTLIERS, LEVERAGE AND INFLUENTIAL POINTS

- *Outlier* - observation with an extreme value on the response variable, y_i
- *Leverage Point* - observation with an extreme value on a predictor variable, x_i
- *Influential Point* - observation that substantially changes the estimate of coefficients after its removal

Multiple Linear Regression

OUTLIERS, LEVERAGE AND INFLUENTIAL POINTS

An *outlier* is an observation that is unusually small or large.

Several possibilities need to be investigated when an outlier is observed:

- There was an error in recording the value.
- The point does not belong in the sample.
- The observation is valid.

How to check:

- Identify outliers from the scatter diagram.
- It is customary to suspect an observation is an outlier if its

$$|\text{standard residual}| > 3.$$

Multiple Linear Regression

Cook's Distance is a measure of an observation or instances' influence on a linear regression.

- Instances with a large influence may be *outliers*, and datasets with a large number of highly influential points might not be suitable for linear regression without further processing such as outlier removal or imputation.
- An observation with *Cook's Distance greater than 1 is worthy of inspection*.
- Each element in the Cook's distance D is the normalized change in the fitted response values due to the deletion of an observation. The Cook's distance of observation i is

$$D_i = \frac{r_i^2}{p \cdot MSE}\left(\frac{h_{ii}}{(1-h_{ii})^2}\right)$$

  where r_i is the i^{th} residual, p is the number of coefficients in the regression model, MSE is the mean squared error, and h_{ii} is the i^{th} leverage value.

Multiple Linear Regression

High Leverage Points

- To quantify an observation's *leverage*, we compute the leverage statistic given by

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{i'=1}^{n}(x_{i'} - \bar{x})^2}$$

- A given observation with leverage statistic that greatly exceeds $2p/n$ has high leverage where p is the number of predictors and n is the sample size.

Outliers

- An observation with *Cook's Distance greater than 1 is worthy of inspection*.
- The Cook's distance of observation i is

$$D_i = \frac{r_i^2}{p \cdot MSE}\left(\frac{h_{ii}}{(1 - h_{ii})^2}\right).$$

Multiple Linear Regression

Influential Points

- Let's identify if there are outliers and leverage points
- The leverage statistic is $2p/n = (2 \times 4)/100 = 0.08$.
- Outliers** ($Y_i > 1$)

```
which(hatvalues(mmodel) > 0.08)
```

```
 4 18 89
 4 18 89
```

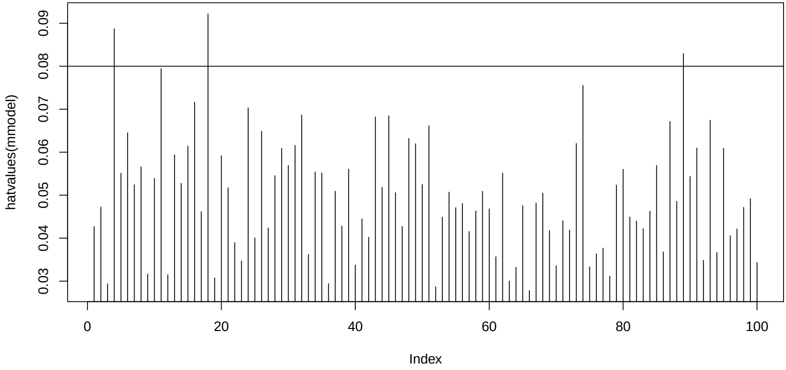
```
which(cooks.distance(mmodel)>1)
```

```
named integer(0)
```

Multiple Linear Regression

High Leverage Points (X_i)

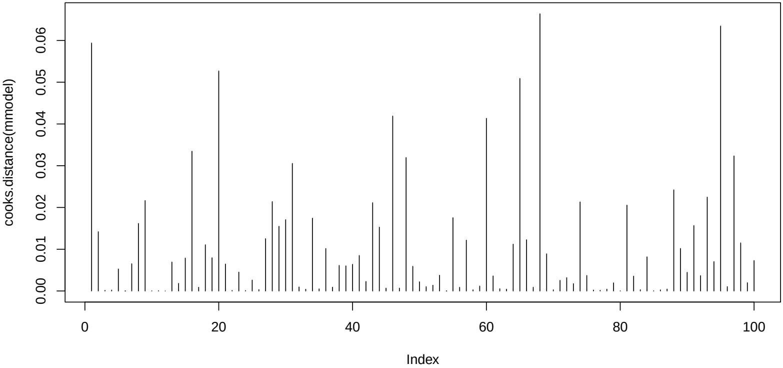
```
plot(hatvalues(mmodel),type="h"); abline(h=0.08)
```



Multiple Linear Regression

High Leverage Points (X_i)

```
plot(cooks.distance(mmodel),type="h"); abline(h=1)
```



DIAGNOSTIC CHECKING

This examines the appropriateness of the model for the data.

Techniques

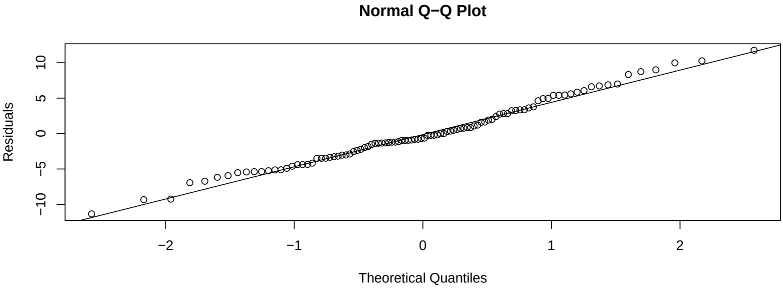
1. Graphical Analysis of Residuals
2. Statistical Tests

Multiple Linear Regression

I. NORMALITY

1. Graphical Analysis of Residuals - Are the residuals normally distributed?

```
qqnorm (residuals(mmodel), ylab="Residuals")
qqline (residuals(mmodel)) #Normality using Q-Q Plots
```



Multiple Linear Regression

I. NORMALITY

2. Statistical Test
   H_o: ϵ's normally distributed.
   H_a: ϵ's not normally distributed.

- Kolmogorov-Smirnov Test
- Shapiro-Wilks Test

The Shapiro-Wilk test is more appropriate method for small sample sizes ($n < 50$ samples) although it can also be handling on larger sample size while Kolmogorov-Smirnov test is used for $n \geq 50$. For both above tests, null hypothesis states that data are taken from normal distributed population. When $p > 0.05$, do not reject the null hypothesis and the residuals/errors are normally distributed.

Source:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6350423/

Multiple Linear Regression

I. NORMALITY

2. Statistical Test

```r
length(residuals(mmodel))
```

```
[1] 100
```

```r
#For demo
shapiro.test(residuals(mmodel))
```

```

	Shapiro-Wilk normality test

data:  residuals(mmodel)
W = 0.98661, p-value = 0.4118
```

Multiple Linear Regression

I. NORMALITY

2. Statistical Test

```r
resid_std <- scale(residuals(mmodel))  # mean=0, sd=1
ks.test(resid_std, "pnorm")
```

```
    Asymptotic one-sample Kolmogorov-Smirnov
    test

data:  resid_std
D = 0.07611, p-value = 0.6085
alternative hypothesis: two-sided
```

Since you do not reject the null hypothesis, so the residuals are normally distributed.

Multiple Linear Regression

Homoscedasticity describes a situation in which the error term (that is, the “noise” or random disturbance in the relationship between the independent variables and the dependent variable) is the same across all values of the independent variables.

Graphs that indicate NON-homoscedasticity (what it should NOT look like)

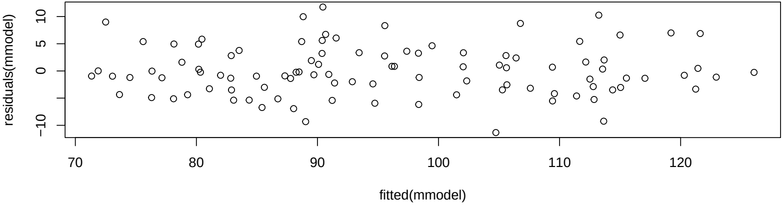
1. Funnel/Cone Shape (Most Common)
2. Inverted Funnel
3. Curved Pattern
4. Bands or Clusters: Residuals appear in horizontal bands
5. Systematic Pattern: Residuals mostly positive on one side and negative on the other.

Multiple Linear Regression

II. HOMOSCEDASTICITY

1. Graphical Analysis of Residuals - the graph should not look like a funnel shape.

```
plot(fitted(mmodel), residuals(mmodel))
```



Homoscedasticity (constant variance) of the errors

- versus the predictions
- versus any independent variable

Multiple Linear Regression

II. HOMOSCEDASTICITY

2. Statistical Test

Hypotheses

- H_o: Heteroscedasticity is not present.
- H_a: Heteroscedasticity is present.

The **Breusch-Pagan-Godfrey Test** (or Breusch-Pagan test) is a test for heteroscedasticity of errors in regression. Homoscedasticity in regression is an important assumption; if the assumption is violated, you won't be able to use regression analysis.

The test statistic for the Breusch-Pagan-Godfrey test is: $n \times R^2$ (with k degrees of freedom), where

- n = sample size
- R^2 = Coefficient of Determination of the regression of squared residuals from the original regression.
- k = number of independent variables.

II. HOMOSCEDASTICITY

2. Statistical Test

```r
library("lmtest")
bptest(mmodel)
```

```
    studentized Breusch-Pagan test

data:  mmodel
BP = 3.4168, df = 4, p-value = 0.4906
```

Thus, the error term is the same across all values of the independent variable using the Breusch-Pagan test.

Heteroscedasticity is not present.

Multiple Linear Regression

II. HOMOSCEDASTICITY

The **Goldfeld Quandt Test** is a test used in regression analysis to test for homoscedasticity. It compares variances of two subgroups; one set of high values and one set of low values. If the *variances differ*, the *test rejects the null hypothesis that the variances of the errors are not constant* (Heteroscedasticity is not present.).

2. Statistical Test

```
gqtest(mmodel, order.by = ScienceScore ~., data = data)
```

```
	Goldfeld-Quandt test

data:  mmodel
GQ = 1.1796, df1 = 45, df2 = 45, p-value =
0.291
alternative hypothesis: variance increases from segment 1 to 2
```

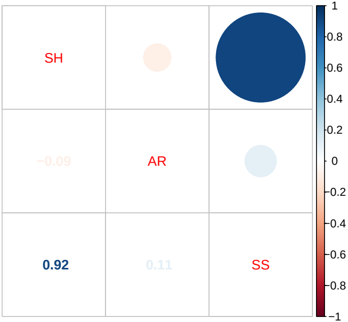
Multiple Linear Regression

III. MULTICOLLINEARITY

This refers to the linear relationship between two or more explanatory variables (predictors).

Example

- Using both cm and m - Perfect multicollinearity
- Height and arm span - Near-perfect multicollinearity

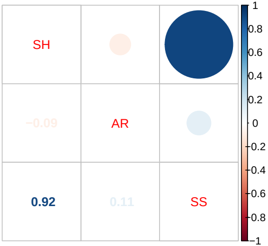


Multiple Linear Regression

III. MULTICOLLINEARITY

EFFECTS OF MULTICOLLINEARITY

- Unstable Regression Coefficients
- Inflated Standard Errors
- Insignificant Predictors Despite High R-squared
- Difficulty in Assessing Individual Effects
- Coefficient Estimates Become Sensitive
- Reduced Statistical Power
- Prediction May Still Be Good



Multiple Linear Regression

III. MULTICOLLINEARITY

2. Statistical Test

```
options(width = 50)
library(car)
vif(mmodel)
```

```
  StudyHours AttendanceRate         Gender
    1.029843       1.040815       1.034448
  SchoolType
    1.018717
```

- VIF $>$ 5 -$>$ moderate concern
- VIF $>$ 10 -$>$ serious multicollinearity

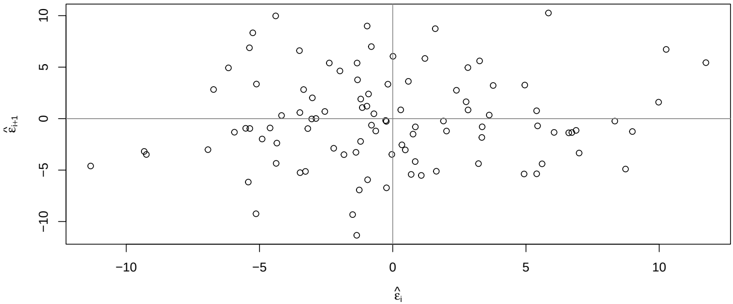
Since there is no high correlation with other independent variables, so it does not have a *multicollinearity issue*.

Multiple Linear Regression

IV. Independence

Autocorrelation and the Durbin-Watson Test

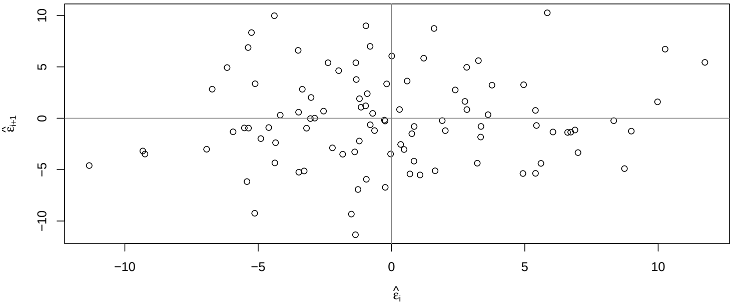
- One of the basic assumptions in the linear regression model is that the random error components or disturbances are identically and independently distributed.
- It is assumed
  that the correlation between the successive disturbances is zero.



Multiple Linear Regression

IV. Independence

1. Graphical Analysis of Residuals



We can see no correlation. If there is, it will follow a linear model.

Multiple Linear Regression

R Codes for the graph

```
n = length(residuals(mmodel))
plot(tail(residuals(mmodel), n-1) ~
       head(residuals(mmodel), n-1),
     xlab=expression(hat(epsilon)[i]),
     ylab=expression(hat(epsilon)[i+1]))
abline(h=0, v=0, col = grey(0.5))
```

Multiple Linear Regression

IV. Independence

2. Statistical Test

Durbin-Watson Test

- H_o: No auto-correlation (the errors are uncorrelated)
- H_a: There exists auto-correlation

```
dwtest(mmodel)
```

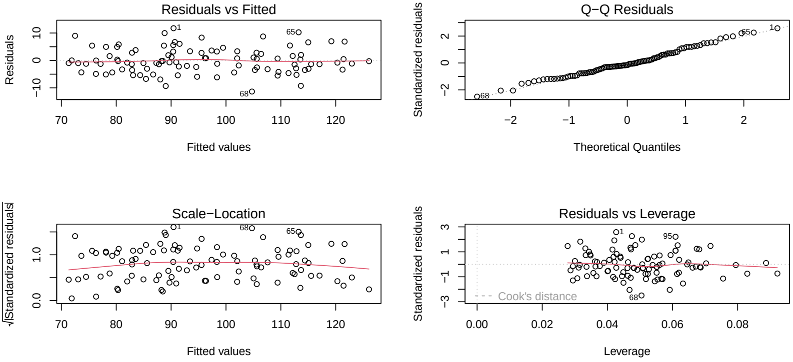
```

	Durbin-Watson test

data:  mmodel
DW = 1.7733, p-value = 0.1216
alternative hypothesis: true autocorrelation is greater than 0
```

The p-value indicates no evidence of serial correlation. Therefore, we can assume that the errors are uncorrelated.

Multiple Linear Regression

Regression diagnostics plots can be created using the R base function *plot()*.

```
par(mfrow = c(2, 2))
plot(mmodel)
```



Multiple Linear Regression

REMEDIAL MEASURES

The assumptions for linear regression are:

1. Linearity: The relationship between X and the mean of Y is linear. (Check Scatterplots)
2. Normality: For any fixed value of X, Y is normally distributed.
3. Homoscedasticity: The variance of residual is the same for any value of X.
4. Multicollinearity: The predictors are not highly correlated.
5. Independence: Observations are independent of each other.

If any of these fail, one needs to do *remedial measures*.

Multiple Linear Regression

REMEDIAL MEASURES

PROBLEM	POSSIBLE REMEDY
Nonlinearity of regression function	- Fit a quadratic or exponential or polynomial regression function - Data transformation
Non-constancy of error variance	- Use weighted least squares - Data transformation
Non-normality of errors	- Data transformation
Non-independence of errors	- Fit time series models

Multiple Linear Regression

There are many ways to transform variables to achieve linearity for regression analysis. Some common methods are summarized below.

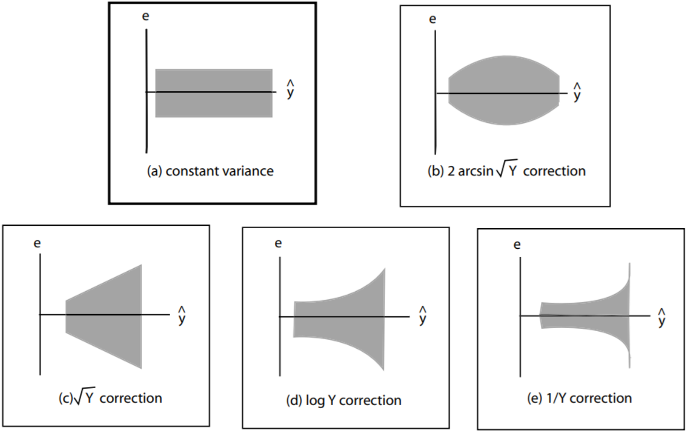
Method	Transform	Regression equation	Predicted value ($\hat{y}$)
Standard linear regression	None	$y = b_0 + b_1 x$	$\hat{y} = b_0 + b_1 x$
Exponential model	$\mathrm{DV} = \log(y)$	$\log(y) = b_0 + b_1 x$	$\hat{y} = 10^{(b_0 + b_1 x)}$
Quadratic model	$\mathrm{DV} = \sqrt{y}$	$\sqrt{y} = b_0 + b_1 x$	$\hat{y} = (b_0 + b_1 x)^2$
Reciprocal model	$\mathrm{DV} = 1/y$	$\frac{1}{y} = b_0 + b_1 x$	$\hat{y} = \frac{1}{b_0 + b_1 x}$
Logarithmic model	$\mathrm{IV} = \log(x)$	$y = b_0 + b_1 \log(x)$	$\hat{y} = b_0 + b_1 \log(x)$
Power model	$\mathrm{DV} = \log(y)$, $\mathrm{IV} = \log(x)$	$\log(y) = b_0 + b_1 \log(x)$	$\hat{y} = 10^{(b_0 + b_1 \log(x))}$

DV = Dependent Variable, IV = Independent Variable

Multiple Linear Regression

Nonconsistancy of error (residual) variance

- residuals versus fitted plots - suggested model corrections



Power Transformations

A *transformation family* is a collection of transformations that are indexed by one or a few parameters that the analyst can select. The family that is used most often is called the **power family**, defined for a strictly positive variable U by

$$\phi(U, \lambda) = U^{\lambda}$$

where $\lambda = 1/2$ or $1/3$ or -1 for inverse, and $\lambda = 1$ is untransformed.

The values of λ used most often in practice are in the range [-1, 1], or less often in the range [-2, 2]. The variable U must be strictly positive for these transformations to be used.

A General Approach to Transformations

The *powerTransform* function in the *car* package is the central tool for helping to choose predictor transformations.

```r
set.seed(123)
n <- 100
data <- data.frame(
  StudyHours = round(runif(n, 2, 20), 1),
  AttendanceRate = round(runif(n, 60, 100), 1),
  Gender = factor(sample(c("Male", "Female"), n, replace = TRUE)),
  SchoolType = factor(sample(c("Public", "Private"), n, replace = TRUE)
)
# Generate Science Score with some realistic relationship
data$ScienceScore <- round(
  40 +
  2.5 * data$StudyHours +
  0.3 * data$AttendanceRate +
  ifelse(data$Gender == "Female", 3, 0) +
  ifelse(data$SchoolType == "Private", 5, 0) +
  rnorm(n, 0, 5),
  1
)
```

A General Approach to Transformations

Test of log transformations

The null hypothesis states that all transformation parameters are equal to zero

$$Ho : \lambda = (0, 0, 0, 0, 0)$$

which corresponds to log transformations for all variables.

Test that no transformations are needed The null hypothesis states that all transformation parameters are equal to one

$$Ho : \lambda = (1, 1, 1, 1, 1)$$

which corresponds to no transformation.

A General Approach to Transformations

```
options(width = 100)
summary(powerTransform(cbind(ScienceScore,
          StudyHours, AttendanceRate, Gender, SchoolType ) ~ 1, data))
```

```
bcPower Transformations to Multinormality
               Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
ScienceScore      0.3317         1.0      -0.4441       1.1074
StudyHours        0.6677         0.5       0.4135       0.9220
AttendanceRate    0.2414         1.0      -1.2453       1.7281
Gender            0.1732         1.0      -0.8067       1.1531
SchoolType        1.2264         1.0       0.2292       2.2236

Likelihood ratio test that transformation parameters are equal to 0
 (all log transformations)
                                 LRT df       pval
LR test, lambda = (0 0 0 0 0) 39.86393  5 1.5908e-07

Likelihood ratio test that no transformations are needed
                                 LRT df     pval
LR test, lambda = (1 1 1 1 1) 10.07499  5 0.073138
```

- Log transformations are not appropriate
- No transformation is statistically acceptable

A General Approach to Transformations

- Discover that data transformation definitely requires a *"trial and error"* approach.
- In building the model, try a transformation and then check to see if the transformation eliminated the problems with the model.
- If it doesn't help, try another transformation, and so on.
- Continue this cyclical process until you have built a model that is appropriate and you can use it. That is, the process of model building includes model formulation, model estimation, and model evaluation:

1. Model building

    - Model formulation
    - Model estimation
    - Model evaluation

2. Model use

Multiple Linear Regression

Variable Selection

- When we fit a multiple regression model, we use the p-value in the ANOVA table to determine whether the model, as a whole, is significant.
- A natural next question to ask is which predictors, among a larger set of all potential predictors, are important.
- We could use the individual p-values and refit the model with only significant terms. But remember that the p-values are adjusted for the other terms in the model.
- So, picking out the subset of significant predictors can be somewhat challenging. This task of identifying the best subset of predictors to include in the model, among all possible subsets of predictors, is referred to as **variable selection**.

Multiple Linear Regression

Variable Selection

For the many possible models, we determine the optimal model based on some criteria: Mallows's C_p, Akaike information criterion (AIC), Bayesian information criterion (BIC), and adjusted R^2.

Mallows's C_p

$$C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2)$$

- The C_p is an estimate of the MSE for a model with d predictors. The C_p adds a penalty to the RSS to adjust for the corresponding decrease in the RSS.
- We choose the model with the lowest C_p value.

Multiple Linear Regression

Variable Selection

For the many possible models, we determine the optimal model based on some criteria: Mallows's C_p, Akaike information criterion (AIC), Bayesian information criterion (BIC), and adjusted R^2.

AIC

$$AIC = \frac{1}{n\hat{\sigma}^2}(RSS + 2d\hat{\sigma}^2)$$

- AIC is proportional to the C_p value.

BIC

$$BIC = \frac{1}{n}(RSS + \log(n)d\hat{\sigma}^2)$$

Like the C_p value and AIC, we select the model with the lowest BIC value.

Multiple Linear Regression

Variable Selection

For the many possible models, we determine the optimal model based on some criteria: Mallows's C_p, Akaike information criterion (AIC), Bayesian information criterion (BIC), and adjusted R^2.

Adjusted R^2

$$\text{Adjusted } R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)}$$

- We select the model with the largest Adjusted R^2.
- The intuition behind the Adjusted R^2 is that once all of the correct variables have been included in the model, adding additional noise variables will lead to only a very small decrease in RSS.
- The Adjusted R^2 statistic pays a price for the inclusion of unnecessary variables.

Multiple Linear Regression

Variable Selection

- Ideally, we try out all possible subset of the predictors. However, trying out every possible subset may be infeasible, even for moderate p.
- We have automated and efficient approaches to choose a smallest set of models to consider using: *Forward selection*, *Backward selection*, *Mixed* or *Stepwise selection*.

Forward selection

- We begin with a null model.
- We then add to the model the variable that results in the lowest RSS.
- Then, we choose the next variable that will result in the lowest RSS for the new two-variable model.
- This is continued until some stopping rule is satisfied.

Variable Selection

Backward selection

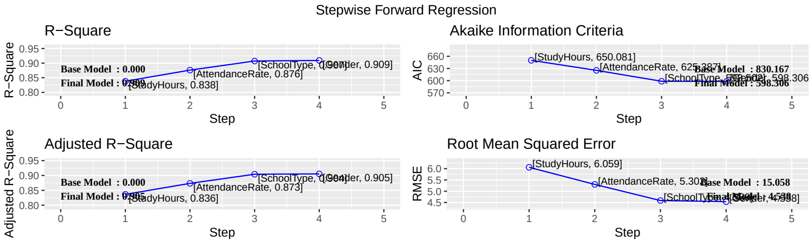
- We start with all variables in the model then remove the variable with the largest p-value, that is, the variable that is the least significant.
- The new $(p-1)$-variable model is fit, and the variable with the largest p-value is removed.
- This continues until all remaining variables have a p-value below some threshold.

Mixed selection

- This is a combination of forward and backward selection.

Forward

```r
library(olsrr)
forward<-ols_step_forward_p(mmodel, details = FALSE)
plot(forward)
```



Multiple Linear Regression

Backward

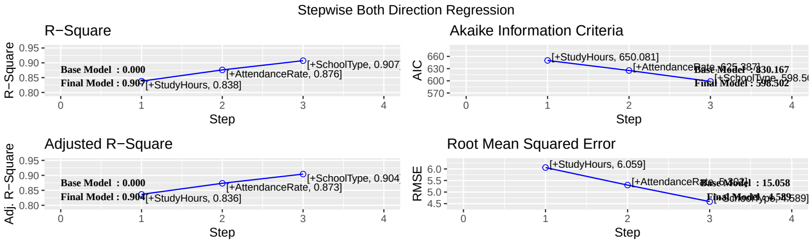
```
ols_step_backward_p(mmodel, details = FALSE)
```

```
[1] "No variables have been removed from the model."
```

Multiple Linear Regression

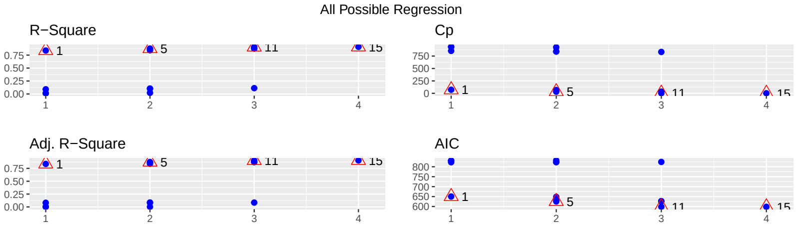
Stepwise

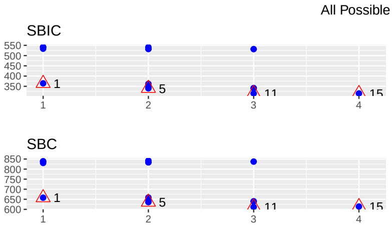
```r
stepwise<-ols_step_both_p(mmodel, details = FALSE,
                          pent=0.05, prem=0.05)
plot(stepwise)
```



Multiple Linear Regression
All Possible

```r
k<-ols_step_all_possible(mmodel)
plot(k)
```





Multiple Linear Regression

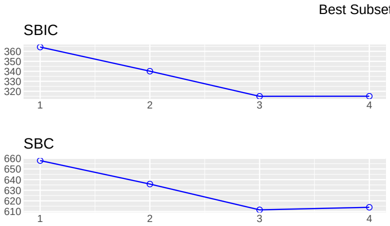
Best Subset

```
k<-ols_step_best_subset(mmodel)
k
```

```
                 Best Subsets Regression
----------------------------------------------------------
Model Index    Predictors
----------------------------------------------------------
     1         StudyHours
     2         StudyHours AttendanceRate
     3         StudyHours AttendanceRate SchoolType
     4         StudyHours AttendanceRate Gender SchoolType
----------------------------------------------------------

                                                     Subsets Regression Summary
-----------------------------------------------------------------------------------------------------------
                       Adj.        Pred
Model    R-Square    R-Square    R-Square     C(p)        AIC         SBIC         SBC          MSEP
-----------------------------------------------------------------------------------------------------------
  1        0.8381      0.8365      0.8314    73.2911    650.0806    364.3206    657.8961    3745.4821     3
  2        0.8760      0.8735      0.8685    35.6297    625.3868    340.1232    635.8075    2897.8683     2
  3        0.9071      0.9042      0.8993     5.1093    598.5023    314.9509    611.5281    2193.7263     2
  4        0.9092      0.9053      0.8994     5.0000    598.3063    315.0393    613.9373    2168.9076     2
-----------------------------------------------------------------------------------------------------------
AIC: Akaike Information Criteria
 SBIC: Sawa's Bayesian Information Criteria
 SBC: Schwarz Bayesian Criteria
 MSEP: Estimated error of prediction, assuming multivariate normality
 FPE: Final Prediction Error
 HSP: Hocking's Sp
 APC: Amemiya Prediction Criteria
```

Multiple Linear Regression

```r
plot(k)
```



Penalized Regression

- Too many predictors $\rightarrow$ overfitting
- Multicollinearity inflates variance

Penalized Objective Function

$$min(RSS + \lambda \cdot Penalty)$$

Ridge Regression (L2): Penalty$=\sum \beta_j^2$

- Shrinks coefficients
- Keeps all variables
- Handles multicollinearity well

LASSO Regression (L1): Penalty$=\sum |\beta_j|$

- Performs feature selection
- Some coefficients become exactly zero

Penalized Regression in R

```
library(glmnet)
```

```
Warning: package 'glmnet' was built under R version 4.5.2
```

```
X <- model.matrix(ScienceScore ~ ., data=data)
y <- data$ScienceScore
ridge <- cv.glmnet(X, y, alpha = 0)
lasso <- cv.glmnet(X, y, alpha = 1)
coef(ridge)
```

```
6 x 1 sparse Matrix of class "dgCMatrix"
                  lambda.1se
(Intercept)       57.0322613
(Intercept)        .
StudyHours         2.2989741
AttendanceRate     0.2140403
GenderMale        -1.4787906
SchoolTypePublic  -5.1285492
```

Penalized Regression in R

```r
coef(lasso, s = "lambda.min")
```

```
6 x 1 sparse Matrix of class "dgCMatrix"
                 lambda.min
(Intercept)      49.3352995
(Intercept)       .
StudyHours        2.6629318
AttendanceRate    0.2602851
GenderMale       -1.2926712
SchoolTypePublic -5.3369628
```

```r
options(width=50)
coef(mmodel)
```

```
     (Intercept)       StudyHours
      48.9516658        2.6724084
  AttendanceRate       GenderMale
       0.2649308       -1.3759026
SchoolTypePublic
      -5.4286289
```

Evaluation of Regression Models

Metric	Formula	Interpretation
MAE	$\frac{1}{n}\sum_{i=1}^{n} \lvert y_i - \hat{y}_i \rvert$	Average absolute error
MSE	$\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$	Penalizes large errors
RMSE	$\sqrt{\text{MSE}}$	Error in original units
R^2	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Goodness of fit

Example

```
set.seed(123)
train_index <- sample(1:nrow(data),
                      size = 0.7 * nrow(data))
train_data <- data[train_index, ]
test_data  <- data[-train_index, ]
```

Model Matrix (Required for glmnet)

```
X_train <- model.matrix(ScienceScore ~ .,
                        train_data)[, -1]
y_train <- train_data$ScienceScore

X_test  <- model.matrix(ScienceScore ~ .,
                        test_data)[, -1]
y_test  <- test_data$ScienceScore
```

Ridge Regression ($\alpha = 0$)

```r
ridge_cv <- cv.glmnet(
  X_train, y_train,
  alpha = 0
)
# Predictions
ridge_pred_train <- predict(ridge_cv, X_train, s = "lambda.min")
ridge_pred_test  <- predict(ridge_cv, X_test,  s = "lambda.min")
# RMSE
ridge_rmse_train <- sqrt(mean((y_train - ridge_pred_train)^2))
ridge_rmse_test  <- sqrt(mean((y_test  - ridge_pred_test)^2))
ridge_rmse_train
```

```
[1] 4.301309
```

```r
ridge_rmse_test
```

```
[1] 5.860556
```

LASSO Regression ($\alpha = 1$)

```
lasso_cv <- cv.glmnet(
  X_train, y_train,
  alpha = 1
)
# Predictions
lasso_pred_train <- predict(lasso_cv, X_train, s = "lambda.min")
lasso_pred_test  <- predict(lasso_cv, X_test,  s = "lambda.min")
# RMSE
lasso_rmse_train <- sqrt(mean((y_train - lasso_pred_train)^2))
lasso_rmse_test  <- sqrt(mean((y_test  - lasso_pred_test)^2))
lasso_rmse_train
```

```
[1] 4.143862
```

```
lasso_rmse_test
```

```
[1] 5.620534
```

RMSE Comparison Table

```r
rmse_results <- data.frame(
  Model = c("Ridge", "LASSO"),
  Train_RMSE = c(ridge_rmse_train, lasso_rmse_train),
  Test_RMSE  = c(ridge_rmse_test,  lasso_rmse_test)
)
rmse_results
```

```
  Model Train_RMSE Test_RMSE
1 Ridge   4.301309  5.860556
2 LASSO   4.143862  5.620534
```

```r
# Using mmodel
lm_pred_train <- predict(mmodel, newdata = train_data)
lm_pred_test <- predict(mmodel, newdata = test_data)
lm_rmse_train <- sqrt(mean((train_data$ScienceScore - lm_pred_train)^2))
lm_rmse_test  <- sqrt(mean((test_data$ScienceScore  - lm_pred_test)^2))
lm_rmse_train
```

```
[1] 4.223739
```

```r
lm_rmse_test
```

```
[1] 5.199311
```

Quantile Regression

This method allows you to analyze the relationship between variables across various quantiles (e.g., median, 25th percentile, 75th percentile) of the dependent variable's distribution, not just the mean.

- Mean regression is sensitive to outliers
- Quantile regression models conditional quantiles

The conditional quantile function of the response variable y given the predictors X is defined as

$$Q_y(\tau \mid X) = X\beta(\tau)$$

where

$$\tau \in (0, 1).$$

Example

```r
library(quantreg)
qr_model <- rq(ScienceScore ~ ., tau = 0.5, data = data)
summary(qr_model)
```

```
Call: rq(formula = ScienceScore ~ ., tau = 0.5, data = data)

tau: [1] 0.5

Coefficients:
                 coefficients lower bd upper bd
(Intercept)      42.80073     40.79253 52.61729
StudyHours        2.71483      2.64353  2.83363
AttendanceRate    0.31064      0.23328  0.34463
GenderMale       -0.00076     -1.28043  0.65306
SchoolTypePublic -4.01577     -5.89045 -2.98003
```

- Estimates median relationship
- Useful for income, wages, and risk analysis

Support Vector Machines (Regression)

Support Vector Regression (SVR) is a powerful machine learning algorithm, an extension of Support Vector Machines (SVMs), used for predicting continuous outputs by finding a function that best fits data within a defined "tube" or margin (the epsilon-insensitive tube).

- Finds a function within an ϵ-insensitive tube

The objective function for Support Vector Regression is:

$$\min \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)$$

Subject to the following constraints:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases}$$

SVR in R

```r
library(e1071)
svr_model <- svm(ScienceScore ~ ., data = data,
                 type = "eps-regression")
summary(svr_model)
```

```
Call:
svm(formula = ScienceScore ~ ., data = data,
    type = "eps-regression")


Parameters:
   SVM-Type:  eps-regression
 SVM-Kernel:  radial
       cost:  1
      gamma:  0.2
    epsilon:  0.1


Number of Support Vectors:  81
```

Interpretation of Parameters

SVM-Type: eps-regression

- Indicates Support Vector Regression
- Errors within $\epsilon = 0.1$ of the predicted value are not penalized
- Focuses on fitting a smooth function rather than minimizing squared error directly

Kernel: Radial (RBF)

- Uses a nonlinear kernel
- Allows the model to capture complex, nonlinear relationships between predictors and ScienceScore
- Suitable when the relationship is not strictly linear

Cost (C = 1)

- Controls the trade-off between model complexity and training error
- A moderate value: Allows some prediction error and Prevents overfitting
- Larger C -> stricter fit
- Smaller C -> smoother model

Interpretation of Parameters

Gamma ($\gamma = 0.2$)

- Determines the influence of individual observations
- Higher γ: Model focuses on nearby points and Risk of overfitting
- Lower γ: Smoother decision function
- $\gamma = 0.2$ suggests moderate nonlinearity

Epsilon ($\epsilon = 0.1$)

- Defines the width of the ϵ-insensitive tube
- Errors smaller than 0.1 are ignored
- Smaller ϵ -> more support vectors, tighter fit

Interpretation of Parameters

Number of Support Vectors: 81

- Out of 100 observations, 81 are support vectors

Indicates:

- Many points lie outside or on the boundary of the ϵ-tube
- The data has complex structure or noise

A large number of support vectors implies:

- Higher model complexity
- Slower prediction compared to simpler models

```r
# Identifying the 81 SV
options(width = 100)
sv_support <- svr_model$SV
head(sv_support)
```

```
  StudyHours AttendanceRate GenderFemale GenderMale SchoolTypePublic
1 -0.7361571      0.3248683            1          0                0
2  1.0184547     -0.6891605            1          0                0
4  1.3498814      1.6705886            0          1                0
5  1.5448382     -0.1205462            1          0                1
6 -1.5939674      1.4241891            1          0                1
7  0.1021574      1.5189582            1          0                0
```

Visualization Idea

```r
# All row indices
all_indices <- 1:nrow(data)
# Support vector indices
sv_index <- svr_model$index
# Non-support vectors
non_sv_index <- setdiff(all_indices, sv_index)
# Check
length(non_sv_index)            # Should be 100 - 81 = 19
```

```
[1] 19
```

```r
non_sv_index                    # Row numbers in 'data'
```

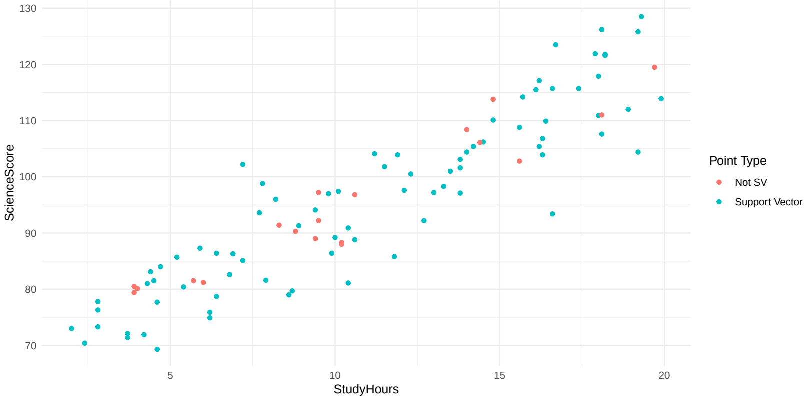
```
 [1]  3 10 12 15 33 42 56 58 59 73 75 76 77 79 80 82 83 85 87
```

```r
# Inspect the non-support vector observations
head(data[non_sv_index, ], 5)
```

```
   StudyHours AttendanceRate Gender SchoolType ScienceScore
3         9.4           79.5 Female     Public         89.0
10       10.2           65.9 Female     Public         88.0
12       10.2           72.0   Male     Public         88.3
15        3.9           88.8 Female    Private         79.4
33       14.4           86.9 Female     Public        106.1
```

Visualization Idea

```
library(ggplot2)
ggplot(data, aes(x = StudyHours, y = ScienceScore)) +
  geom_point(aes(color = ifelse(1:nrow(data) %in%
  sv_index, "Support Vector", "Not SV"))) +
  labs(color = "Point Type") + theme_minimal()
```



Regression Activity

Data Code

```r
set.seed(123) # Use your set.seed
n <- 100
data <- data.frame(
  StudyHours = round(runif(n, 2, 20), 1),
  AttendanceRate = round(runif(n, 60, 100), 1),
  Gender = factor(sample(c("Male", "Female"), n,
                         replace = TRUE)),
  SchoolType = factor(sample(c("Public", "Private"), n, replace = TRUE)
)
# Generate Science Score with some realistic relationship
data$ScienceScore <- round(
  40 +
  2.5 * data$StudyHours +
  0.3 * data$AttendanceRate +
  ifelse(data$Gender == "Female", 3, 0) +
  ifelse(data$SchoolType == "Private", 5, 0) +
  rnorm(n, 0, 5),
  1
)
```

Regression Activity

- Fit SVR
- Identify support vectors (SVs)
- Fit Linear Regression, Ridge, and LASSO only using the SV observations
- Compute Train and Test RMSE for all models
- Display results interactively in Shiny

Submission Requirement: Knit your RMarkdown file to HTML and upload the resulting file to Canvas. Use the same title file.

References

- Downey, A. B. (2014). Think Stats: Exploratory Data Analysis (2nd ed.). O'Reilly Media / Green Tea Press.
- Grolemund, G., & Wickham, H. (2017). R for data science: Import, tidy, transform, visualize, and model data. O'Reilly Media.
- Wickham, H., & Grolemund, G. (2023). R for Data Science: Import, Tidy, Transform, Visualize, and Model Data (2nd ed.)
- James, G. et al., An Introduction to Statistical Learning, 2021
- Tan, Steinbach, Kumar, Introduction to Data Mining, 2022
- R Documentation: kmeans, dbscan, hclust, arules

THANK YOU!